ALEC L. CRITTEN

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Citizenship: United States of America
Languages: English, French (∼A1/A2)

EDUCATION

Swansea University *September 2023 - present*
PhD, Computer Science *Swansea, Wales, United Kingdom*
Thesis: "*SMT Solving for Graph-Theoretic Representations of Railway Systems*"

University of Vermont *August 2018 - May 2021*
B.S. in Mathematics, minor in Computer Science *Burlington, Vermont, United States*
Honors College Scholar
Final GPA: 3.62/4.0
Thesis: "*Characterizing Insecure Error Distributions for Various RLWE Problems*"

RESEARCH INTERESTS

My work centres around the applications of formal methods to real-world systems, particularly through verification using automated and interactive theorem proving. As a PhD student and member of the Swansea Centre for Research in Digital Railways, I have worked to ensure the correctness of the design of safety-critical railway systems using techniques such as SMT solving.

TECHNICAL ABILITIES

Programming Languages	Python, C++, Visual Basic, C#, Haskell, Prolog, CSP, Java, SageMath, MATLAB
Proof Assistants and Theorem Provers	Z3, cvc5, Rocq/Coq, Lean
Operating Systems	Windows, Linux (Ubuntu)
Additional Skills	Git, Jira, Bitbucket, LaTeX, Wolfram Mathematica

WORK EXPERIENCE

Swansea University October 2023 - *present*
Graduate Teaching Assistant *Swansea, Wales, UK*

- Responsible for supporting the teaching mission of the University's Computer Science Department. Provides support through lab demonstrations, assisting lecturers, and marking courseworks and examinations.
- Obtained the Advance HE Associate Fellowship (AFHEA) teaching qualification as part of my position.
- Modules supported: CSCM985 (Logic, Modelling, and Verification), CSCM/385 (Modelling and Verification Techniques), CSC375 (Logic for Computer Science), CSCM/364 (Software Testing), CSCM/313 (High-Integrity and Critical Systems), CS-205 (Declarative Programming), CS-135 (Professional Issues 2: Software Development), CSC005 (Computational Probability)

Siemens July 2025 - September 2025
Software Engineer (internship) *Chippenham, England, UK*

- Engaged with the formal methods team in the R&D division of Siemens Mobility, particularly with regard to integrating graph-theoretic SMT solving into current internal railway verification software.

Peraton September 2022 - August 2023
Applications Programmer and Analyst *Warner Robins, Georgia, U.S.*

· Responsible for providing software engineering support for a program under the U.S. Air Force. Maintained and developed code for a GUI project.

deciBel Research August 2021 - August 2022
Software Developer *Huntsville, Alabama, U.S.*

· Responsible for software design, development, test and integration support for a modeling and simulation effort in U.S. ballistic missile defense.

Georgia Military College June 2021 - July 2021
Adjunct Faculty and Tutor in Mathematics *Warner Robins, Georgia, U.S.*

· Adjunct faculty member, lecturer (College Prep Algebra II), and tutor in mathematics.

Polymath REU, virtual Summer 2020
Undergraduate Researcher and Group Leader *Virtual*

· Participated in a multi-national collaborative mathematical research experience. Led small research group in number theory. Developed mathematical conjectures and proofs. Contributed to published writing, available at: https://arxiv.org/abs/2010.14932

CONFERENCES AND EVENTS

UKRRIN Student Conference 2026 January 2026
Swansea, Wales, UK

Co-chair of conference, member of programme committee and organising committee

Summer School Marktoberdorf 2025 August 2025
Marktoberdorf, Germany

Attendee of summer school

BCTCS 2025 April 2025
Glasgow, Scotland, UK

Presented a talk: "Developing user propagators for graph-based SMT reasoning"

UKRRIN Student Conference 2025 January 2025
Sheffield, England, UK

Gave a poster presentation: "SMT solving for graph-theoretic representations of railway systems"

Proof and Computation Autumn School 2024 September 2024
Fischbachau, Germany

Attendee of autumn school

SMT/SAT/AR Summer School 2024 June 2024
Nancy, France

Attendee of summer school